

LOGO / SITE TITLE

Blender Animation Tutorial

Home

Tutorial Steps

Resources

Download

Large Example Image

Learn to Create a Blender Animation

• Intro Text Goes Here.

• Intro Text Goes Here.

• Intro Text Goes Here.

About This Tutorial

Description Text Here.

Screenshot Image

Blender Basics

Interface Image

• Info Text Here.

Download Blender

Download Addon

Links & Requirements Text

Example Gallery

Example Image

Example Image

Example Image

Caption Text

Caption Text

Caption Text

Ready to Start the Tutorial?

Start Tutorial Button

Step 1

Step 2

Step 3

Resources

Contact | Credits | Blender Website

LOGO / SITE TITLE

Blender Animation Tutorial

Home

Tutorial Steps

Resources

Download

Step # - [Step Title Here]

Brief description or overview text goes here.

Section Title

Step 1:

Instructional text goes here. This is a step-by-step explanation of what to do in this step.

Large Example Image

Step 2:

Large Example Image

• Instructional text goes here. This is another step-by-step explanation of what to do in this step.

Step 3:

Instructional text goes here. This is yet another step-by-step explanation of what to do in this step.

Large Example Image

Step 3:

Instructional text goes here. This is yet another step-by-step explanation of what to do in this step.

Large Example Image

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1C0B19

Midnight Violet

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140D4F

Deep Twilight

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7EBCE6

Maya Blue

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17EC7E

Spring Green

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04F749

Electric Green

Home page content:



Logo.

Header:

Kerbal Space Program Blender Tutorial

Home. Add-on, Importing, Texturing, Rigs.

Banner image:



Learn how to create Kerbal Space Program animations in Blender

- Get Kerbal Rigs
- Import Ships
- Texturing
-

About This Tutorial

In this tutorial we will be going in depth in a step by step process teaching you how to create kerbal space program animations in blender. A basic understanding of blender, as well as owning a copy of KSP on your computer, is necessary for this tutorial. Watch the video version here: https://youtu.be/Qd0wp3JavXc?si=_ipIVaSMH79PRKD7

Screenshot image:



Blender

Interface image:



We will be working with Blender 3.6 or earlier. It is necessary to import ships and parts in this version of Blender, as the importer addon will not work in other versions. We will also need to download the io_object_mu addon from GitHub.

Link 1: <https://www.blender.org/download/lts/3-6/> Link 2:
https://github.com/taniwha/io_object_mu

Animation Examples

Check out some of my past animations here:



Images are links.

https://youtu.be/i_Skofc0U8g?si=u4zFidci3ogBcWRr
https://youtu.be/_qyqJ2rNQ1c?si=YcxIPUmhTHYyq0fp
https://youtu.be/hfMcnV5JGHs?si=M2n3jYE5_wY8dSN-

Ready to start the tutorial?

Start Here

Links to the first page

Step 1

Setting up addon

Step 2

Importing ships and parts

Step 3

Texturing

Step 4

Kerbal Rigs

Page 1 content

Step 1 – Installing the Blender Add-on

Overview:

Before we can import Kerbal Space Program ships into Blender, we first need to install the KSP Blender add-on. This add-on allows Blender to read craft files and individual game parts so they can be edited, rendered, or animated.

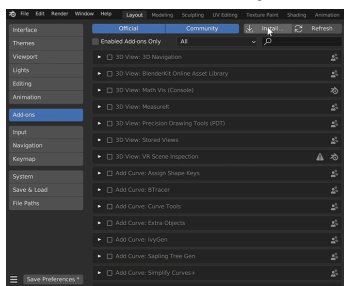
Installing the Add-on in Blender

Open the Add-on Installer

Once the add-on ZIP file has finished downloading, open Blender.

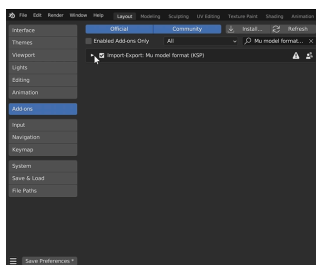
To install the add-on:

- Click Edit in the top menu.
- Select Preferences.
- Navigate to the Add-ons tab.
- Click Install.
- Select io_object_mu-master.zip in downloads



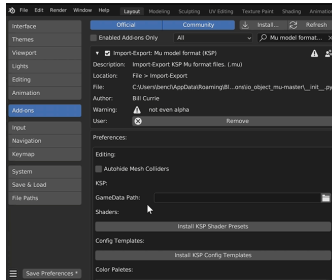
After clicking install, locate the downloaded add-on ZIP file and select it.

Once installed, make sure the add-on checkbox is enabled so Blender activates it.



Before closing the Preferences window, expand the newly installed add-on settings. Inside the panel you will see an option labeled GameData Path.

Blender needs to know where your Kerbal Space Program game files are located.

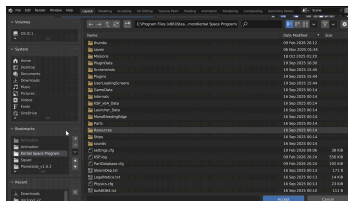


If you installed the game through Steam, you can find it at:

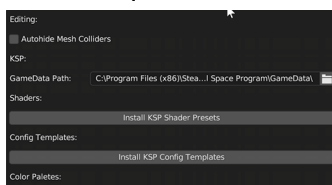
OS
Program Files (x86)
Steam
SteamApps
Common
Kerbal Space Program

Once you locate the folder, click the plus icon to bookmark it. This will make it easier to return to later.

Select the GameData folder and click Accept.



Make sure the final folder listed in the path is GameData.
This is required for the add-on to work correctly.



Overview

Once the add-on is installed and connected to your Kerbal Space Program files, you can begin importing ships and individual parts into Blender. In this section, you will learn how to import a full craft file, clean up the imported objects, and prepare the ship so it can be animated.

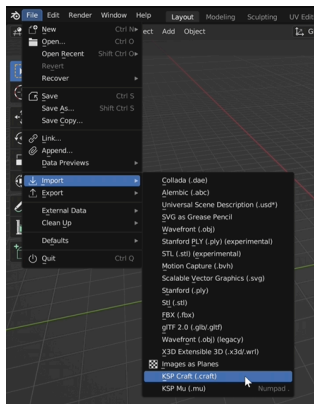
Importing a Craft File

Open the Craft Importer

Instruction Text

To import a ship from Kerbal Space Program:

1. Click **File** in the Blender menu.
2. Select **Import**.
3. Choose **KSP Craft**.



Next, navigate to the **Kerbal Space Program** folder that you bookmarked earlier when installing the add-on.

From here you have two options:

- **Ships** – Contains the stock ships included with the game.
- **Saves** – Contains ships you have built and saved in your own game files.

Name	Date Modified
thumbs	09 Feb 2025 20:12
Saves	09 Sep 2025 01:18
Mission	18 Oct 2025 01:20
PluginData	19 Sep 2025 16:30
Screenshots	19 Sep 2025 15:45
Plugins	19 Sep 2025 15:44
UserLoadingScreens	19 Sep 2025 15:44
GameData	16 Sep 2025 00:14
Internal	16 Sep 2025 00:14
KSP_Job_Data	16 Sep 2025 00:14
Launcher_Data	16 Sep 2025 00:14
ModBlendingEdge	16 Sep 2025 00:14
Parts	16 Sep 2025 00:14
Resources	16 Sep 2025 00:14
Ships	16 Sep 2025 00:14
sounds	16 Sep 2025 00:14

For this tutorial we will import a **stock ship**.

Navigate to:

Ships

VAB

Kerbal X

Select the craft file and import it into Blender.

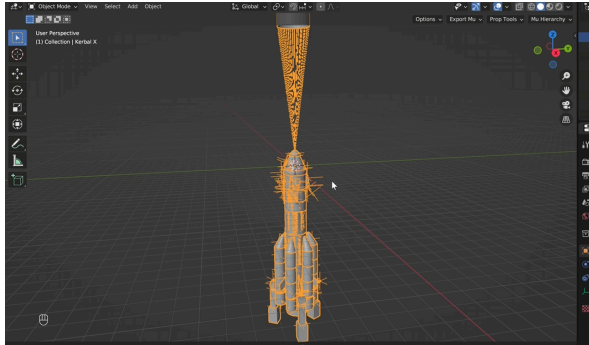
Name	Date Modified	Size	File Properties
Jumping Piss.craft	31 Jan 2024 00:39	14 KB	
Heater & Spacecraft.craft	20 Jan 2025 23:14	24 KB	
Kerbal 1.craft	01 Oct 2024 15:00	43 KB	
PT Series Munlander.craft	04 Sep 2024 23:16	65 KB	
Comet 1.craft	04 Sep 2024 23:16	31 KB	
Launcher A1.craft	04 Sep 2024 23:07	130 KB	
2-MHP Satellite Launch Kit.craft	17 Jul 2024 22:53	112 KB	
Box-Powered Space Probe.craft	17 Jul 2024 12:24	14 KB	
GC-V3.craft	13 Jul 2024 18:33	45 KB	
Dynawing.craft	10 Jul 2024 18:33	27 KB	
Aerodyne.craft	13 Jul 2024 18:32	52 KB	
Kerbal X.craft	10 Jul 2024 15:42	119 KB	
Kerbal 3.craft	10 Jul 2024 15:42	116 KB	
Veneratic Survey Satellite.craft	10 Jul 2024 15:42	31 KB	
Kerbal 1.craft	10 Jul 2024 15:42	51 KB	
Kerbal 2.craft	10 Jul 2024 15:42	29 KB	
Orbiter 1A.craft	10 Jul 2024 15:42	49 KB	
Space Station Core.craft	10 Jul 2024 15:42	38 KB	
Orbiter One.craft	10 Jul 2024 15:42	34 KB	
Science 1.craft	10 Jul 2024 15:42	52 KB	
Star Shredder.craft	10 Jul 2024 15:42	39 KB	
Super-Heavy Lander.craft	10 Jul 2024 15:42	61 KB	
Two-Stage Lander.craft	10 Jul 2024 15:42	24 KB	
Kerbal X.craft			Import

Blender Version Note

Working in Different Blender Versions

At this point, you do **not need to remain in Blender 3.6**. You can save the imported craft file and then append it into another Blender version if you prefer.

This tutorial continues using **Blender 3.6**, since it is commonly used for animation workflows, but the process should work in other compatible versions as well.



Separating the Ship Parts

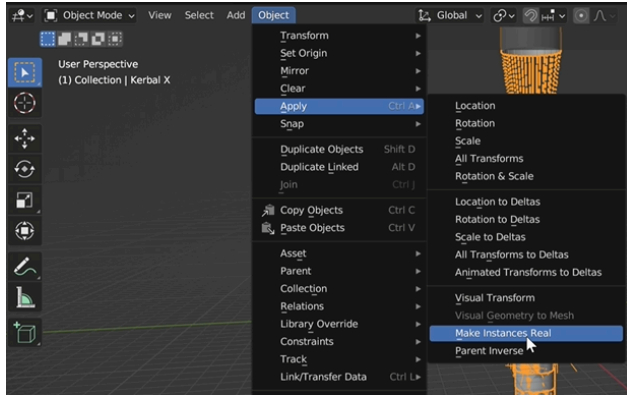
Converting Instances into Real Objects

When the craft first imports into Blender, it will appear as **one combined object** and will contain many **empty objects** used by the game.

To separate the parts:

1. Select the imported ship.
2. Click **Object** in the top menu.
3. Choose **Apply**.
4. Select **Make Instances Real**.

This converts the instanced parts into individual objects so they can be edited and animated.



Removing Unnecessary Empty Objects

Cleaning Up the Scene

After separating the objects, the scene will still contain many **empty objects** that are not needed for animation.

Instead of deleting them one by one, you can remove them quickly using the following method:

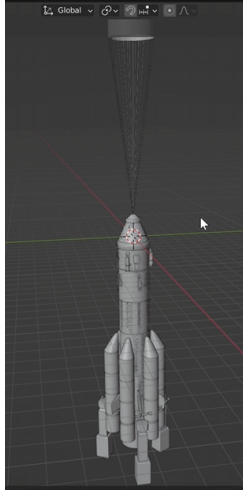
1. Click **Select**
2. Choose **Select by Type**
3. Select **Mesh**

This selects all mesh objects in the scene.

Next:

1. Click **Select** again
2. Choose **Invert Selection**
3. Press **X** to delete the selected objects

This removes all the empty objects and leaves only the ship parts.



Checking Materials and Parts

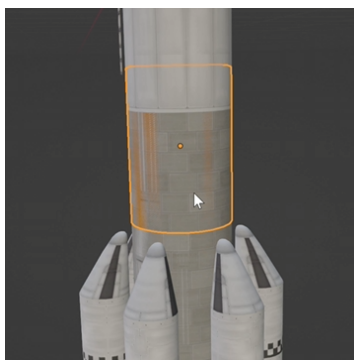
Reviewing Imported Materials

Switch to **Material Preview Mode** to see how the craft materials were imported.

You should now see:

- Colored parts
- Engine covers
- Various materials used by the craft

At this stage, you want to **delete any parts you do not need** and clean up the model before continuing.



Preparing the Ship for Animation

Parenting the Craft to an Empty Object

Since the craft is made up of many individual parts, animating it directly can be difficult. A simple solution is to parent the ship to a single control object.

To do this:

1. Press **Shift + A**
2. Add an **Empty** → **Cube**
3. Move the empty object to the **center of mass** of the ship
4. Press **A** to select all objects
5. Press **Ctrl + P**
6. Choose **Parent to Object**

Now when you **move or rotate the empty object**, the entire ship will move together as a single unit.



Result

At this point you should have:

- A clean ship model
- Unnecessary objects removed
- The entire craft controlled by a single empty object

Your ship is now ready to be **positioned, animated, and rendered inside Blender.**

Step 3 – Changing and Editing Part Textures

Overview

Sometimes when a craft is imported into Blender, certain parts may use textures that you do not want. For example, booster cones or fuel tanks might import with a color or material different from what you expected.

In this section, you will learn how to open the shading editor, change the texture used by a part, and create unique textures for individual pieces.

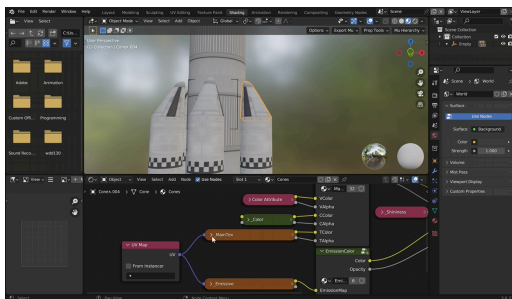
Opening the Shading Workspace

Access the Texture Controls

To begin editing textures, switch to Blender's **Shading workspace**.

1. Click the **Shading** tab at the top of Blender.
2. Select the **part of the ship** you want to modify.

This workspace allows you to view and edit the **materials and textures** used by each object.



Changing a Part's Texture

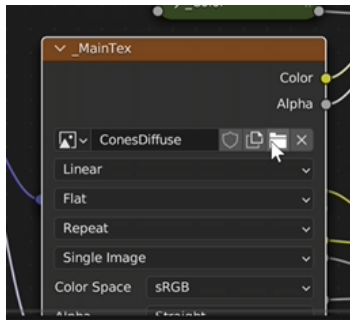
Selecting a Different Texture

Once you have selected a part, locate the **Image Texture node** in the shader editor.

You should see a texture labeled **MainTex**. This is the main texture used by the object.

To change it:

1. Click the **Image Texture node** labeled “MainTex.”
2. Click the **folder icon** next to the texture name.

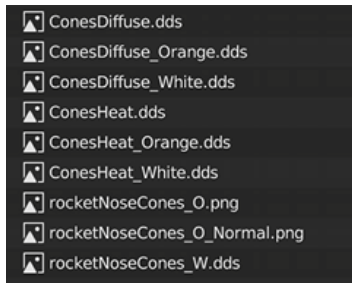


This will open a list of available textures for that part.

For example, a cone may have multiple texture options such as:

- Regular
- Orange
- White

Select the texture you want to use. In this example, we will choose the **white** version.



Adjusting Normal Maps

Matching the Texture's Surface Detail

Some parts also include **bump or normal maps** that control how surface details appear in lighting.

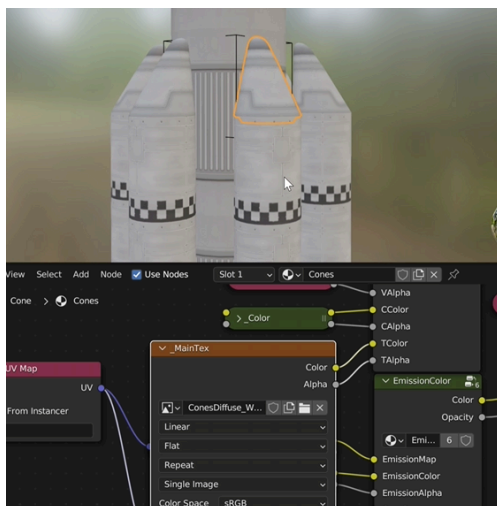
If you change the main texture, you should also update the **normal map** to match it.

For example:

- If you selected the **white texture**, choose the **white normal map** as well.

Matching these ensures the surface details appear correctly when the object is rendered.

Image Placeholder



Changing Textures on Individual Parts

Editing One Part Without Affecting Others

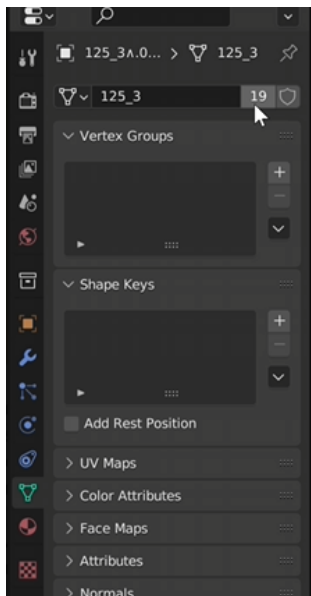
Many imported ship parts share the same material. Because of this, changing the texture on one object may also change it on every identical part.

If you want to modify **only one object**, you will need to create a separate material.

To do this:

1. Open the **Material Properties** panel.
2. Locate the **material name** and the number next to it.
3. Click the **number button** to make the material single-user.

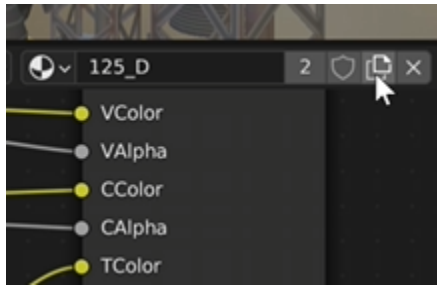
This disconnects the object from the shared material.



Next:

1. Click the **Duplicate Texture** button in the shader editor.
2. Change the texture as needed.

Now the texture will apply **only to that object**, without affecting the others.



Result

After adjusting the textures, your craft parts should now display the **colors and materials you want**.

You can now:

- Customize individual ship components
- Match textures across parts
- Prepare the craft for **lighting and rendering**

Step 4 – Using Kerbal Character Rigs

Overview

To bring life to your scenes and animations, you can add **Kerbal character rigs** to your project. These rigs allow you to pose and animate Kerbals with fully controllable bodies, faces, and equipment.

The rigs used in this tutorial were originally created by **Peter Edwardson** on the Kerbal Space Program forums. Because they were created independently from the game, they are **free to download and share**.

The versions provided in this tutorial have also been modified to include additional features such as improved facial controls and alternate suit colors.

Downloading the Kerbal Rigs

Getting the Rig Files

Instruction Text

Before using the Kerbal characters in Blender, you will need to download the rig files.

The rigs are available through the **download link provided with this tutorial**. Two versions of the rig are included.

Download the files and save them somewhere easy to access on your computer. You will import or append these rigs into your Blender project later when setting up your scene.

Image Placeholder

Large screenshot showing:

- Download page or file folder containing the rigs
-

Features of the Kerbal Rigs

What the Rigs Include

Instruction Text

These rigs are designed specifically for animation and include several built-in features that make posing and animating Kerbals easier.

Some of the included features are:

- **Fully rigged bodies** for posing and movement
- **Facial controls** for expressions and animation
- **Jetpack rigs** for EVA-style scenes
- **Suit variations**, including the classic orange Kerbal suit
- Additional controls such as **eyelids** for more expressive characters

These features make the rigs flexible enough for a wide range of animation styles.

Image Placeholder

Large screenshot showing:

- Kerbal rig controls
- Armature visible in Blender

Modified Versions Used in This Tutorial

Additional Improvements

The rigs used in this tutorial are slightly modified versions of the originals.

The changes include:

- Improved eyelid controls
- Alternate suit color variants
- Minor adjustments to make animation easier

Kerbal rig link

<https://drive.google.com/drive/folders/1W28PFKNJGkFn8t1PLQYGzsy3hURXcpPt?usp=sharing>

These additions help make the characters more expressive and easier to work with when creating animated scenes.



Using Kerbal Rigs in Your Animations

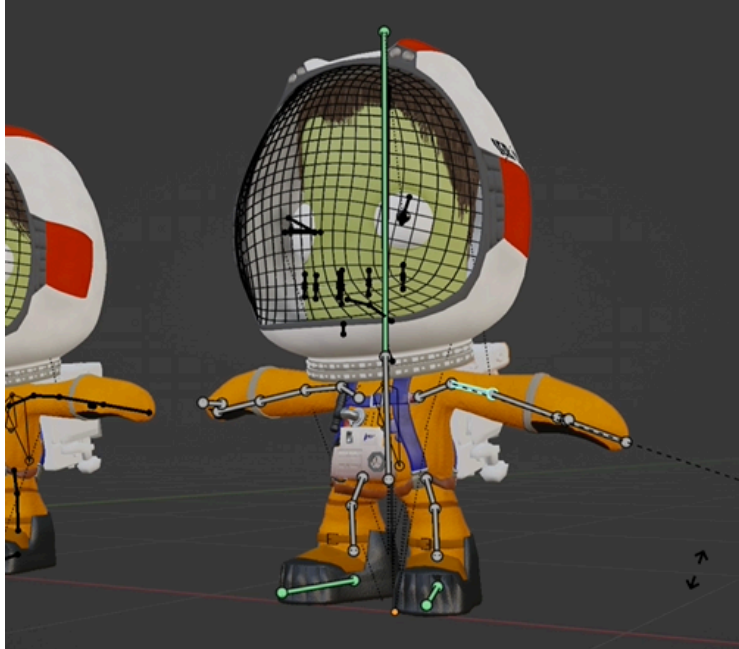
Preparing for Animation

Once downloaded, the Kerbal rigs can be added to your Blender project alongside your imported spacecraft.

Using these rigs, you can:

- Pose Kerbals inside or outside ships
- Animate EVA scenes
- Create cinematic moments involving characters

Combined with the ships and environments you imported earlier, the rigs allow you to create **complete Kerbal Space Program animation scenes** inside Blender.



Result

At this point in the tutorial you should now have:

- Imported **ships and parts from Kerbal Space Program**
- Cleaned and organized your craft models
- Adjusted **textures and materials**
- Downloaded **Kerbal animation rigs**

You now have everything needed to begin **building scenes and creating animations** in Blender.